

Computational Physics_ Dr. Ejtehad

Homework02_Deposition

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1- Random Ballistic Deposition

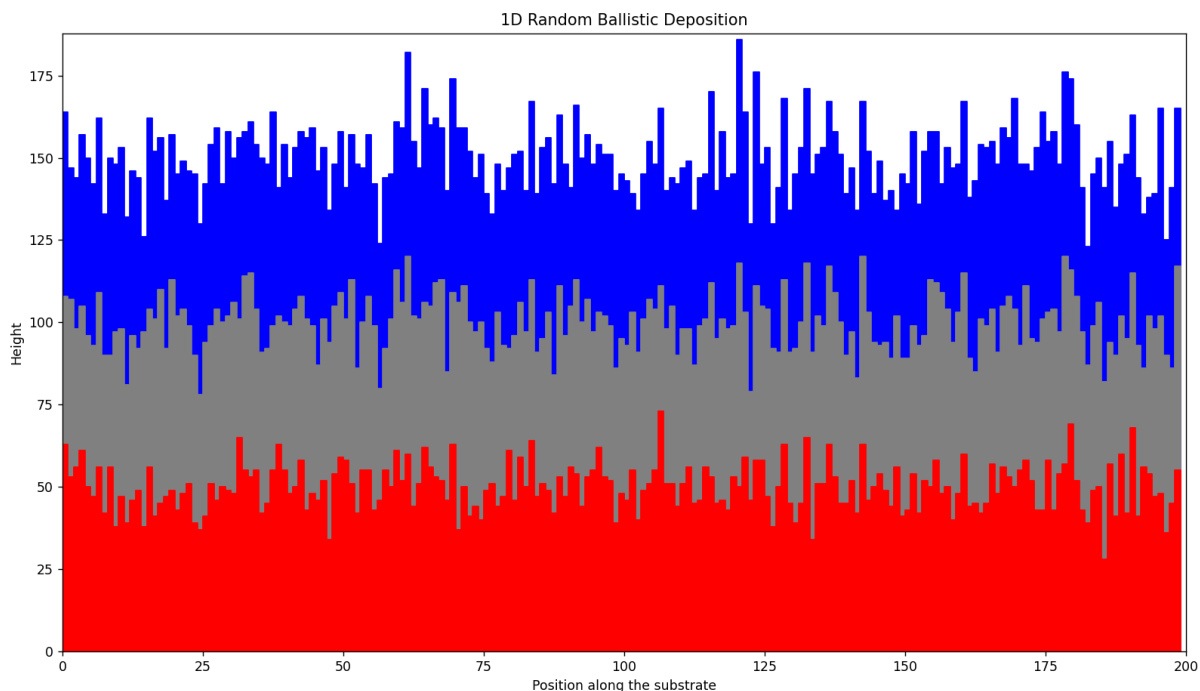
General Idea

This script simulates 1D random ballistic deposition, a model used to study surface growth and roughening. In this process, particles are deposited randomly onto a 1D substrate of length L , one at a time, and they stick to the highest available position at that site. Over time, this leads to an evolving surface with a characteristic roughness.

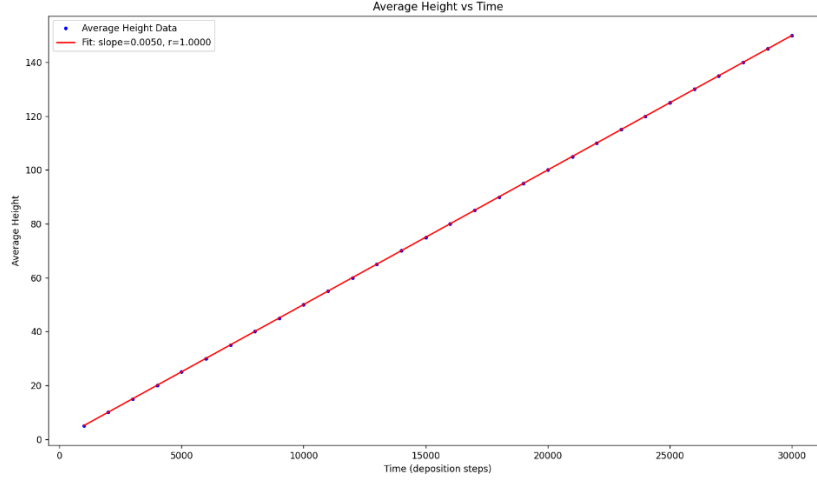
Algorithm and Approach

To implement this in Python, we define a 1D array representing the substrate, initially set to zero. At each deposition step, a random site is selected, and its height is increased by one. This process is repeated for multiple layers, with each layer consisting of $50L$ deposition steps to ensure uniform coverage. At regular intervals, we record the average height and roughness, storing this data for later analysis.

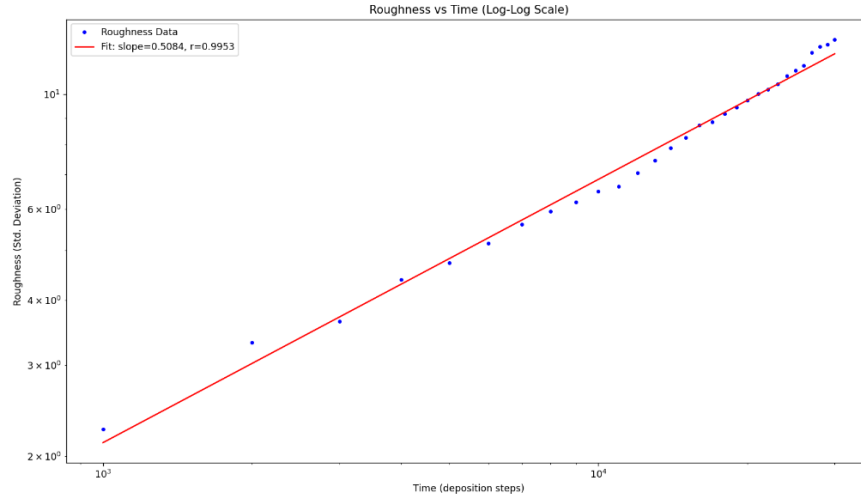
Results and Visualisation



Analysis



The average height vs. time plot shows a clear linear relationship, which aligns with the expectation that height should grow proportionally with deposition time. The linear regression fit in red confirms this, with a slope indicating a steady growth rate. The correlation coefficient $r=1.0000$ suggests an almost perfect linear relationship, meaning that each deposition step contributes consistently to the height increase.



In the log-log plot of roughness vs. time, we observe a linear relationship, indicating a power-law scaling behaviour. This suggests that the roughness $W(t)$, defined as the standard deviation of the height distribution, follows the scaling law:

$$W(t) \sim t^\beta$$

After five tests, the average measured slope (β) was **0.509**, closely matching the theoretical value of $\beta = 0.5$. The relative error is **1.8%**, indicating high accuracy with only minor deviations due to statistical fluctuations. The strong agreement confirms that the system correctly models surface growth dynamics

2- Ballistic Deposition with Relaxation

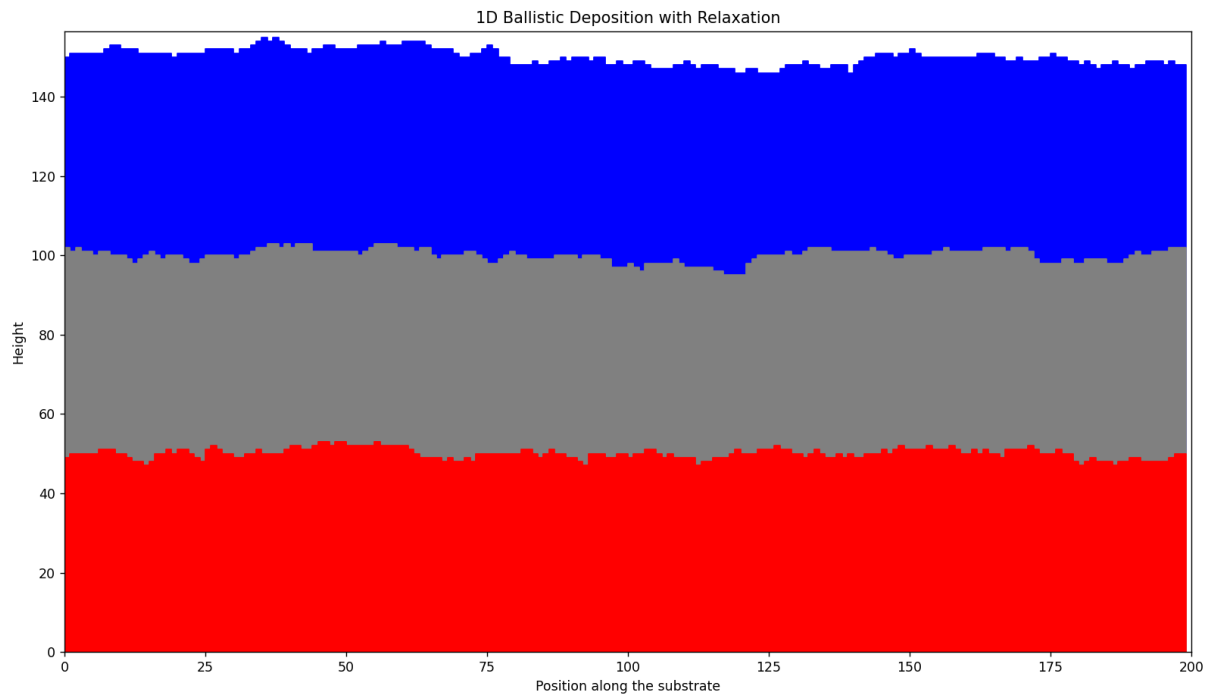
General Idea

This simulation extends 1D ballistic deposition by introducing a relaxation mechanism, where deposited particles can move to a neighbouring site if it results in a lower height. This creates a smoother surface compared to standard ballistic deposition, as the system tends to minimize height differences between neighbouring columns. By tracking the average height and roughness, we can analyse how this relaxation affects surface growth over time.

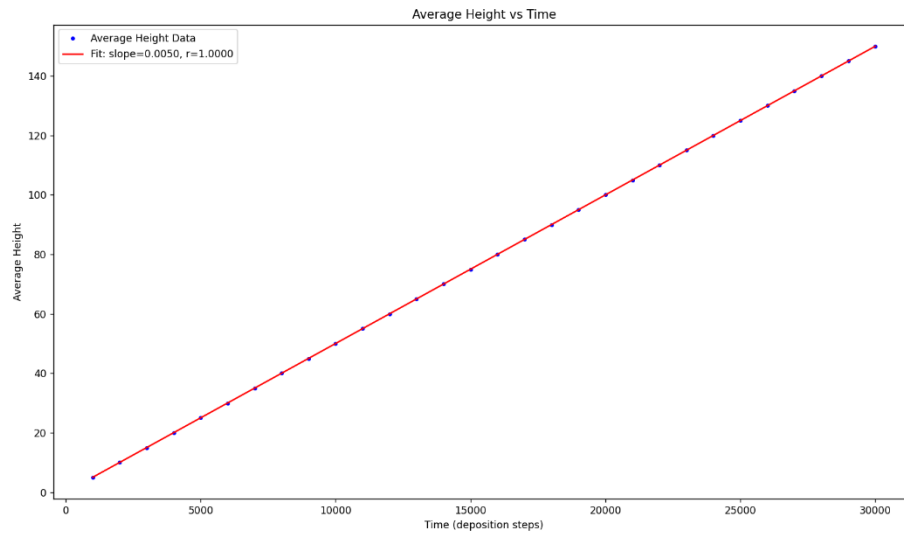
Algorithm and Approach

To implement this in Python, we define a 1D array representing the substrate, initially set to zero. At each deposition step, a random site is selected, and its neighbouring heights are checked. If the site is taller than both neighbours, the particle is deposited at the original site. Otherwise, it moves to the shorter neighbour. If both neighbours are of the same height, the deposition site is chosen randomly between them. This process is repeated for multiple layers, each consisting of 50L deposition steps to ensure uniform coverage. At regular intervals, the average height and roughness are recorded for later analysis.

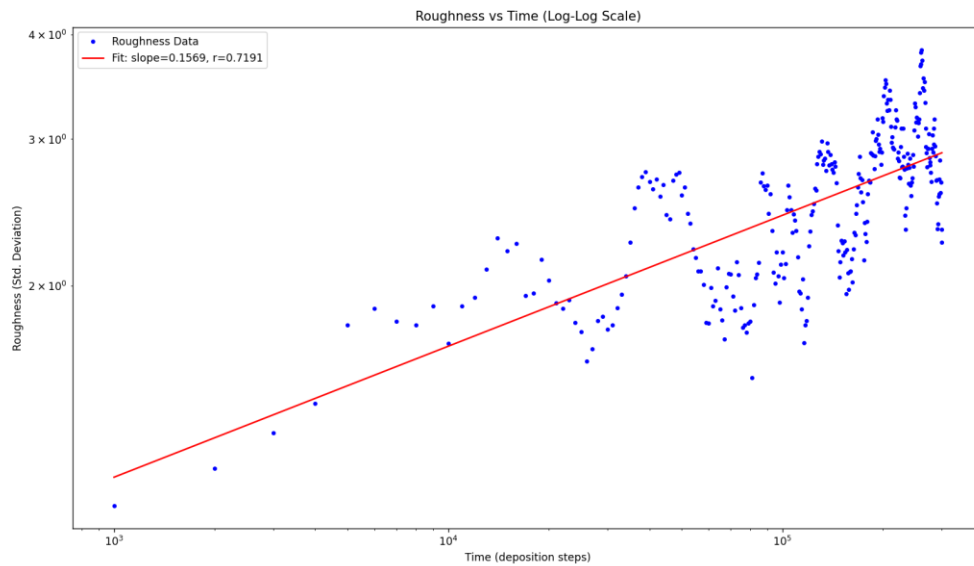
Results and Visualisation



Analysis



The results confirm that ballistic deposition with relaxation leads to a linear increase in average height over time, as shown in the average height vs. time plot. This behaviour is expected since each deposition step contributes consistently to surface growth, with minor fluctuations due to random site selection.



However, the roughness vs. time plot on a log-log scale reveals an important phenomenon: while roughness initially follows a power-law growth, it eventually saturates and no longer increases at the same rate. This occurs because correlations develop between neighbouring columns over time. Initially, height fluctuations are independent, but as deposition progresses, the surface becomes more correlated, meaning that neighbouring sites influence each other's growth. This limits the extent to which roughness can continue increasing, leading to a saturation region where the roughness stabilises.

$$W_s \sim t_s^\beta \sim L^{z\beta} \sim L^\alpha$$

Using another Python script to analyse the data and test different system sizes ($L = 50, 100, 200, 400$), we obtained the following values: Growth exponent: 0.2154, Roughness exponent α : 0.6517, Dynamic exponent z : 1.0, Scaling relation check: 0.2154, Number of deposited particles needed for saturation ($L = 200$): 300000.

3- Side Deposition

General Idea

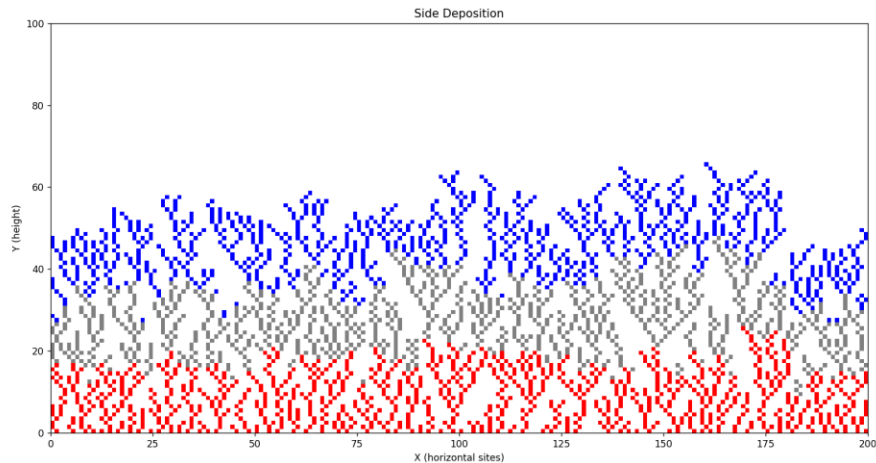
This script simulates 2D side deposition on a grid of size $H \times L$. Particles are dropped from the top at a random x -position and move downward until they either reach the bottom or encounter a neighbouring particle. By grouping particles into three batches (each batch having a distinct colour), we can visually track how the deposition layers build up over time.

Algorithm and Approach

To implement this in Python, we create a 2D NumPy array representing the deposition grid, initially filled with zeros. Each falling particle is assigned a batch ID based on its index. It then descends from the top row ($y = H - 1$) at a random x -position until one of two conditions is met: 1- It reaches $y=0$ (the bottom), 2- It detects a neighbour (below, below-left, below-right, left, or right) that has already been deposited.

When either condition is satisfied, the particle is placed at its current position, and its batch ID is stored in the grid. After all particles have been deposited, the grid is visualised with a custom colour map that highlights the different batches.

Results and Visualisation



4- Side Deposition 1 Point

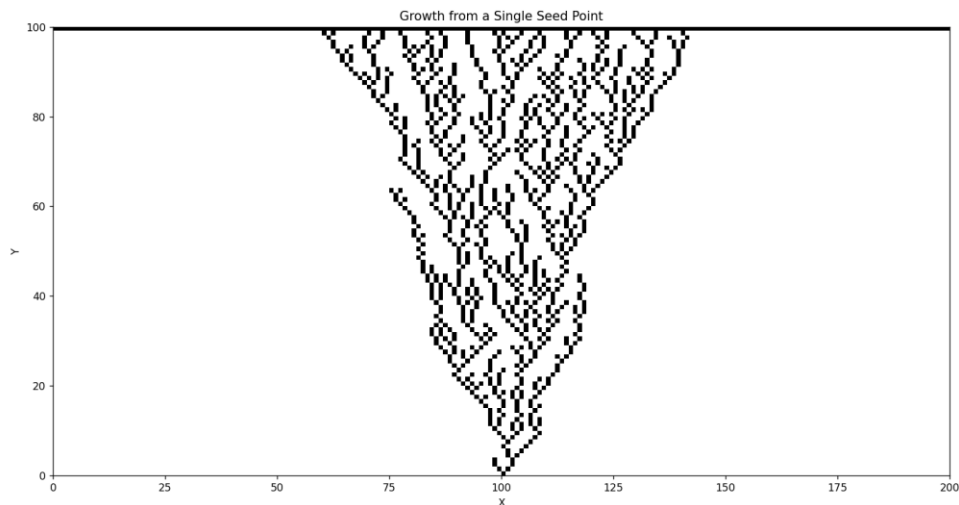
General Idea

This script simulates 2D growth from a single seed point in the bottom centre of a grid. Particles are introduced from the top at random horizontal positions and move downward. When a particle detects an occupied neighbour (including the cell itself), it stops and occupies that position. Over time, a cluster grows upward from the seed.

Algorithm and Approach

To implement this in Python, we define a 2D NumPy array of size $H \times L$ initialised with zeros. A single seed is placed at $(x=L/2, y=0)$. We then drop N particles from random x -positions at the top row ($y=H-1$). Each particle moves down one cell at a time. At each step, we check if the particle or any of its immediate neighbours is occupied. If so, the particle settles in its current position and the loop ends. Otherwise, it continues descending until it reaches $y=-1$ (i.e., it has left the grid). This process yields a growing cluster anchored at the seed.

Results and Visualisation



Analysis

The resulting cluster exhibits a branch-like structure emanating from the single seed. Because each particle stops as soon as it encounters an occupied cell or neighbour, the cluster spreads upward and outward over time. The density and shape of the cluster depend on the randomness of horizontal positions and the local neighbourhood checks.

5- Angled Ballistic Deposition

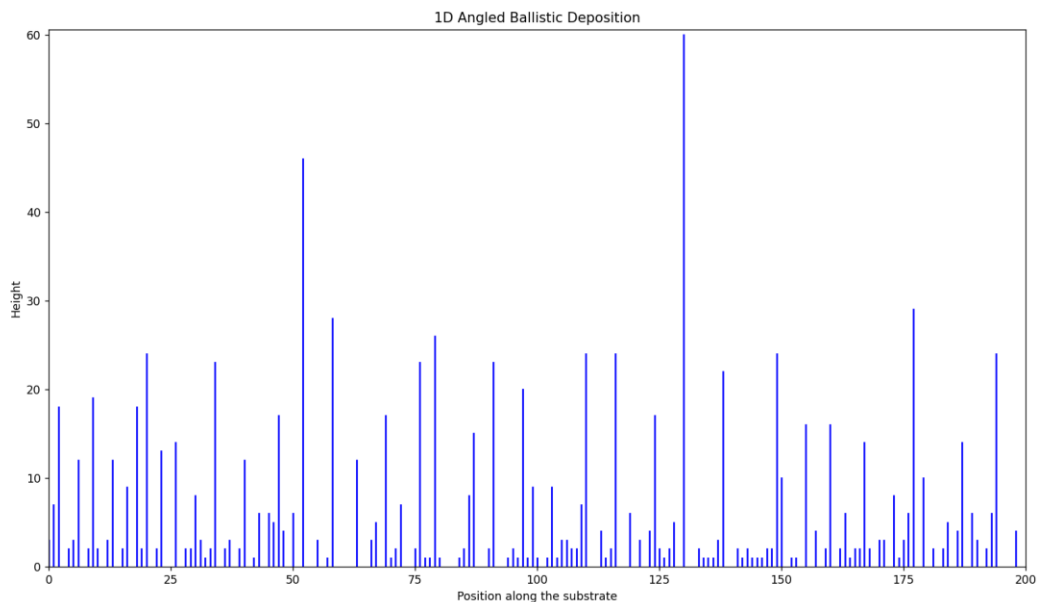
General Idea

This script simulates 1D ballistic deposition at an inclined angle of 60 degrees from the vertical. Particles enter from a randomly selected column at the top and travel horizontally and vertically according to the chosen angle. If a particle encounters a column that is already tall enough to block its path, it settles on that column and increases its height by one unit. Over multiple deposition events, a characteristic surface profile emerges, which can be analysed by computing the average height and roughness (standard deviation of heights).

Algorithm and Approach

We maintain a 1D array of column heights. Each new particle appears at a random column, then travels diagonally left according to the deposition angle. If the particle's effective height is blocked by an existing column, it collides and increments that column's height. After N depositions, the resulting surface profile is plotted, and we compute the mean height and standard deviation (roughness) to quantify surface growth.

Results and Visualisation



Analysis

Because columns can “shadow” those behind them, taller columns tend to keep growing while shorter columns receive fewer particles. This creates needle-like peaks and amplifies height differences, reflecting the competitive nature of angled deposition. Over time, the profile becomes increasingly uneven, with the roughness capturing these growing height disparities.